

Having fun with RISC-V: Data Annex

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TABLE I: Full breakdown behind Table II of the letter. Compute window (the measurement loop), both compilers. IPC is instret/cycles. Ratios are GHC/MicroHs; whole is the whole-run cycle ratio including RTS start-up.

Program	MicroHs (RV32-xfun)					GHC (RV32)			GHC/MicroHs		
	cyc	instret	IPC	combi	fn.y	cyc	instret	IPC	cyc	instret	whole
flite											
Permsort	36 051	27 142	0.75	11 758	321	697 202	545 378	0.78	19.34	20.09	34.45
Queens2	49 257	38 521	0.78	15 162	1 121	787 836	634 362	0.81	15.99	16.47	33.28
Taut	103 472	82 268	0.80	29 945	1 382	1 354 694	1 079 880	0.80	13.09	13.13	30.12
Ordlist	32 764	24 526	0.75	13 488	419	385 074	309 580	0.80	11.75	12.62	33.77
Braun	170 346	113 383	0.67	54 570	260	1 978 346	1 626 875	0.82	11.61	14.35	25.35
Sumpuz	28 192	20 902	0.74	6 616	650	304 720	235 986	0.77	10.81	11.29	34.13
Sudoku	1 426 304	920 354	0.65	250 579	9 337	12 519 991	10 626 218	0.85	8.78	11.55	12.35
TreeSum	620 772	417 892	0.67	114 681	24 573	4 270 049	3 608 820	0.85	6.88	8.64	14.44
TreePari	108 292	85 839	0.79	30 449	1 024	720 013	608 424	0.85	6.65	7.09	26.24
Adjoxo	64 202	46 064	0.72	18 921	341	399 772	321 945	0.81	6.23	6.99	31.04
Mss	670 984	569 051	0.85	70 794	12 430	3 484 015	2 992 377	0.86	5.19	5.26	12.59
Knuthbendix	81 353	54 625	0.67	19 924	1 314	347 140	262 239	0.76	4.27	4.80	29.00
Countdown	67 582	42 914	0.63	14 759	1 140	282 910	221 114	0.78	4.19	5.15	29.81
Parts	20 254 455	14 298 995	0.71	1 331 971	108 940	77 238 658	62 811 959	0.81	3.81	4.39	4.15
Whillex	228 910	98 883	0.43	47 613	123	807 645	700 703	0.87	3.53	7.09	19.34
Clausify	565 127	181 571	0.32	92 060	3 423	1 865 500	1 577 713	0.85	3.30	8.69	12.64
Queens	206 259	156 086	0.76	48 190	5 307	575 472	486 475	0.85	2.79	3.12	19.86
TribeLie	38 055	24 577	0.65	11 482	513	104 307	85 367	0.82	2.74	3.47	31.33
Fibonacci	9 427	6 154	0.65	1 149	429	16 709	13 739	0.82	1.77	2.23	35.62
Factorial	4 339	2 228	0.51	325	160	5 718	4 201	0.73	1.32	1.89	36.67
SumEuler	123 920	63 406	0.51	16 721	1 997	100 946	54 584	0.54	0.81	0.86	23.21
nofib-imaginary											
Paraffins	92 251	66 056	0.72	20 232	2 446	—	—	—	—	—	28.28
WheelSieve2	8 690	5 314	0.61	1 415	199	81 100	59 182	0.73	9.33	11.14	35.97
WheelSieve1	5 574	3 306	0.59	809	110	43 840	29 503	0.67	7.87	8.92	35.73
Primes	12 451	5 819	0.47	1 788	182	75 633	55 602	0.74	6.07	9.56	34.73
Kahan	18 451	9 688	0.53	1 662	483	82 300	63 982	0.78	4.46	6.60	34.74
Integrate	191 417	86 968	0.45	11 222	3 517	720 908	556 639	0.77	3.77	6.40	—
NQueens	53 041	39 385	0.74	11 991	1 133	124 867	105 770	0.85	2.35	2.69	29.43
GenRegexps	9 405 962	2 184 980	0.23	706 654	104 443	18 860 261	16 399 199	0.87	2.01	7.51	2.75
Tak	13 387	9 756	0.73	3 077	219	19 965	15 275	0.77	1.49	1.57	33.88
RFib	47 918	21 078	0.44	2 151	572	39 484	23 762	0.60	0.82	1.13	30.06
X2n1	86 576	36 844	0.43	4 298	2 208	48 760	25 388	0.52	0.56	0.69	26.15
nofib-spectral											
Atom	15 906	9 271	0.58	1 784	250	—	—	—	—	—	—
Life	24 018	17 185	0.72	5 855	463	267 312	217 716	0.81	11.13	12.67	—
Constraints	213 600	153 491	0.72	58 336	2 900	2 302 187	1 875 718	0.81	10.78	12.22	23.46
Cse	68 514	47 010	0.69	15 711	1 045	717 091	584 623	0.82	10.47	12.44	29.72
Lcss	55 276	45 663	0.83	8 888	332	454 307	369 461	0.81	8.22	8.09	—
Ansi	233 123	127 465	0.55	36 514	2 384	1 905 310	1 594 433	0.84	8.17	12.51	—
Calendar	1 232 767	487 357	0.40	174 031	13 530	10 004 626	8 238 879	0.82	8.12	16.91	—
Scs	16 289	11 388	0.70	4 122	103	114 698	91 428	0.80	7.04	8.03	35.17
Awards	1 443 731	681 289	0.47	225 844	21 145	10 148 588	8 181 794	0.81	7.03	12.01	—
Multiplier	3 896 127	2 008 814	0.52	558 207	25 938	25 594 685	21 202 391	0.83	6.57	10.55	—
PrettyN	19 389	9 969	0.51	3 426	272	114 615	87 248	0.76	5.91	8.75	34.74

Program	MicroHs (RV32-xfun)					GHC (RV32)			GHC/MicroHs		
	cyc	instret	IPC	combi	fn.y	cyc	instret	IPC	cyc	instret	whole
Fish	2 950 673	1 062 583	0.36	310 733	53 643	16 457 961	13 571 170	0.82	5.58	12.77	7.65
Sorting	51 202	34 135	0.67	10 110	330	278 299	221 038	0.79	5.44	6.48	30.49
Minimax	5 825 338	2 732 263	0.47	920 934	51 823	27 242 534	23 274 663	0.85	4.68	8.52	–
Cicelli	688 113	558 300	0.81	117 843	8 186	3 048 618	2 641 114	0.87	4.43	4.73	–
Lambda	113 851	81 234	0.71	31 140	761	431 833	366 292	0.85	3.79	4.51	24.43
Grep	8 987 266	3 086 321	0.34	1 039 603	73 953	24 272 685	20 390 923	0.84	2.70	6.61	–
Cryptarithm1	2 525 294	1 103 980	0.44	304 118	68 281	6 518 309	5 581 419	0.86	2.58	5.06	–
Mandel	394 335	169 825	0.43	19 417	6 926	984 597	815 349	0.83	2.50	4.80	14.33
LastPiece	1 290 419	693 983	0.54	177 979	10 049	3 088 574	2 620 849	0.85	2.39	3.78	–
Knights	11 507 455	6 739 535	0.59	1 834 309	108 095	19 776 993	17 083 661	0.86	1.72	2.53	–
Sphere	947 908	409 729	0.43	63 614	15 289	1 506 832	1 171 404	0.78	1.59	2.86	8.04
Treejoin	9 754 215	4 015 715	0.41	393 636	16 739	–	–	–	–	–	1.72
Mandel2	1 306 881	688 565	0.53	108 985	25 200	423 131	233 854	0.55	0.32	0.34	5.37
nofib-real											
Infer	3 425 601	1 623 776	0.47	587 696	10 921	13 900 962	11 297 129	0.81	4.06	6.96	–
Prolog	2 892 573	1 215 716	0.42	488 928	17 570	–	–	–	–	–	6.25
Parser	3 374 090	1 350 785	0.40	504 990	4 908	4 293 325	3 387 281	0.79	1.27	2.51	3.32
Compress	30 696 709	12 992 239	0.42	3 348 657	10 371	30 619 690	26 514 548	0.87	1.00	2.04	–
nofib-shootout											
NBody	3 595 635	2 047 290	0.57	240 404	33 911	26 381 366	21 523 172	0.82	7.34	10.51	8.85
Fft	207 393	112 843	0.54	9 480	2 225	1 092 785	924 601	0.85	5.27	8.19	21.05
SpecNorm	1 151 262	514 536	0.45	64 591	26 343	2 667 349	2 049 845	0.77	2.32	3.98	–
nofib-hartel											
Wang	138 325	64 129	0.46	16 087	1 913	813 684	645 608	0.79	5.88	10.07	24.37
Typecheck	22 538 279	13 793 125	0.61	3 196 890	171 699	130 178 479	105 205 480	0.81	5.78	7.63	6.05
Event	791 358	446 175	0.56	151 924	9 911	2 918 611	2 468 005	0.85	3.69	5.53	10.78
Sched	12 466 176	4 949 458	0.40	1 492 684	7 845	–	–	–	–	–	1.58
gadt											
RoseTree	33 459	23 270	0.70	6 125	1 429	492 759	418 951	0.85	14.73	18.00	33.49
Queue	97 201	63 094	0.65	18 740	2 472	–	–	–	–	–	28.72
MergeSort	250 912	174 331	0.69	45 191	4 987	2 893 846	2 362 773	0.82	11.53	13.55	–
InsertSort	233 133	157 503	0.68	42 269	1 918	2 675 734	2 288 109	0.86	11.48	14.53	22.84
BTree	83 230	59 342	0.71	17 353	1 617	946 727	799 381	0.84	11.37	13.47	28.63
Zipper	59 105	42 431	0.72	13 197	1 290	648 926	534 251	0.82	10.98	12.59	31.01
HeapSort	193 873	121 884	0.63	33 949	2 845	–	–	–	–	–	–
QuadTree	343 117	188 715	0.55	57 557	6 768	2 509 122	2 112 789	0.84	7.31	11.20	18.42
Patricia	101 047	74 244	0.73	15 923	3 535	600 964	508 469	0.85	5.95	6.85	26.68
List	534	139	0.26	2	0	–	–	–	–	–	38.36
Tuple	9 379	4 199	0.45	821	297	1 205	756	0.63	0.13	0.18	35.49
nofib-gc											
Power	2 631 871	1 031 446	0.39	321 235	15 764	5 177 280	4 256 151	0.82	1.97	4.13	4.57
Circsim	18 610 231	10 959 739	0.59	1 646 158	67 330	32 290 938	26 666 641	0.83	1.74	2.43	–

TABLE II: Cache behaviour. MicroHs counters are for the compute window. The GHC counters are emitted once per run by the harness and therefore cover the whole run, RTS start-up included; they are not directly comparable with the MicroHs columns and no ratio is given.

Program	MicroHs (RV32-xfun), compute window						GHC (RV32), whole run				
	i-miss	d-miss	fills	miss stall	RAM rd	RAM wr	i-miss	d-miss	fills	miss stall	
flite											
Permsort	327	1	20	706	416	1 994	326	1 129	1 418	25 721	
Queens2	442	1	27	906	576	9 791	393	1 100	1 473	27 088	
Taut	464	2	32	1 064	640	48 306	408	1 096	1 482	27 033	
Ordlist	356	1	17	693	368	537	268	510	770	13 950	
Braun	1 168	1	291	8 302	4 816	107 823	430	2 524	2 922	53 080	
Sumpuz	293	1	50	1 208	945	5	454	219	620	10 942	

Program	MicroHs (RV32-xfun), compute window						GHC (RV32), whole run			
	i-miss	d-miss	fills	miss stall	RAM rd	RAM wr	i-miss	d-miss	fills	miss stall
Sudoku	12 831	20	6 009	190 805	97 936	1 050 625	2 254	14 572	16 500	306 253
TreeSum	4 470	4	18	4 904	480	457 850	136	4 704	4 829	87 286
TreePari	982	1	12	1 227	336	48 611	88	604	684	12 233
Adjoxo	974	1	50	1 923	944	23 594	343	219	547	9 639
Mss	4 199	2	233	9 654	14 224	331 612	283	1 926	2 195	39 846
Knuthbendix	1 036	3	366	7 814	6 464	31 120	1 443	472	1 821	32 090
Countdown	999	1	53	1 939	960	16 672	455	488	899	16 231
Parts	56 381	118 077	124 697	2 565 232	2 042 272	8 626 772	2 475	87 950	90 185	1 708 622
Whilex	2 086	1	79	3 212	1 360	168 426	465	1 136	1 568	28 810
Clausify	3 775	1	61	5 022	1 104	448 085	457	1 942	2 384	43 655
Queens	3 241	1	29	3 853	560	129 278	148	582	723	12 967
TribeLie	752	1	27	1 310	2 384	4 222	146	222	360	6 342
Fibonacci	48	1	9	215	241	2	21	7	26	422
Factorial	7	1	8	160	609	226	19	15	34	604
SumEuler	416	1	21	788	624	39 157	40	11	50	823
nofib-imaginary										
Paraffins	533	1	58	1 549	1 600	35 833	956	1 438	2 335	42 596
WheelSieve2	120	1	47	951	849	3	576	151	696	12 288
WheelSieve1	78	1	41	820	753	3	480	91	555	9 911
Primes	100	1	18	400	401	2	136	103	227	4 100
Kahan	187	1	31	707	785	41	528	168	641	11 300
Integrate	701	1	38	1 365	833	51 983	848	817	1 589	28 646
NQueens	566	1	23	968	448	7 211	109	68	168	3 113
GenRegexps	218 063	627	98 315	2 540 312	2 372 881	5 503 842	863	31 375	32 150	602 049
Tak	120	1	11	324	401	3	21	6	26	495
RFib	241	1	16	495	433	2	361	65	384	6 691
X2n1	386	1	46	1 185	1 377	396	563	110	621	10 721
nofib-spectral										
Atom	64	1	32	622	961	198	847	1 500	2 300	41 345
Life	207	2	54	1 158	1 345	244	485	396	868	15 563
Constraints	1 651	3	88	3 316	1 600	139 433	1 137	3 352	4 428	79 683
Cse	623	1	53	1 528	4 272	19 035	626	793	1 383	25 056
Lcss	209	1	45	1 027	913	15	539	721	1 205	21 648
Ansi	2 594	1	175	5 868	24 048	140 255	713	4 070	4 747	85 965
Calendar	6 590	2	4 059	137 381	110 608	901 229	1 625	16 427	17 933	334 598
Scc	234	1	48	1 068	1 089	71	776	207	933	16 525
Awards	10 002	40	538	25 600	18 944	1 159 776	1 449	12 593	13 896	256 903
Multiplier	32 796	28	13 292	409 301	293 985	2 915 285	3 255	30 560	33 576	631 878
PrettyN	297	1	113	2 345	3 441	1 388	950	298	1 172	20 705
Fish	29 211	4	16 163	486 105	374 096	2 033 858	4 282	26 810	30 876	580 848
Sorting	477	1	147	3 161	4 144	6 943	1 251	494	1 654	29 220
Minimax	33 609	3	1 895	89 277	34 160	4 831 365	3 549	32 663	36 028	681 275
Cichelli	3 371	1	403	14 263	10 224	496 904	899	2 355	3 186	57 377
Lambda	1 562	1	88	3 441	1 632	60 276	418	453	836	15 098
Grep	103 925	1	42 143	1 322 279	1 207 056	6 410 635	1 739	29 364	30 934	579 687
Cryptarithm1	19 479	1	123	21 693	2 096	2 087 257	686	9 697	10 332	188 982
Mandel	1 417	1	55	2 369	4 368	113 926	907	888	1 714	30 626
LastPiece	6 159	4	1 159	40 235	32 400	1 008 166	1 166	2 933	4 040	73 466
Knights	55 071	4 949	5 089	182 079	87 824	9 120 952	2 803	10 215	12 785	236 851
Sphere	4 886	6	587	16 989	9 904	383 256	2 842	1 173	3 747	66 887
Treejoin	209 156	82	228 798	4 688 625	3 684 736	2 330 492	1 209	15 340	16 460	303 795
Mandel2	9 986	4	199	14 587	3 328	642 405	607	225	758	13 300
nofib-real										
Infer	18 949	19	3 931	136 627	76 432	2 795 880	5 011	11 433	16 083	297 136
Prolog	20 251	36	5 470	196 825	121 025	2 289 075	4 555	13 460	17 749	331 800
Parser	32 262	3	4 066	157 419	93 856	2 749 978	2 829	4 950	7 615	137 790
Compress	181 052	32 027	42 249	1 148 886	843 712	20 820 292	1 504	34 710	36 098	677 245
nofib-shootout										
NBody	66 271	10	22 626	525 408	369 569	1 900 436	8 496	61 568	69 474	1 320 633
Fft	724	12	89	2 271	2 065	52 487	1 273	1 672	2 871	51 635
SpecNorm	12 476	620	131	20 162	60 672	530 655	1 175	3 902	4 994	90 994

Program	MicroHs (RV32-xfun), compute window						GHC (RV32), whole run			
	i-miss	d-miss	fills	miss stall	RAM rd	RAM wr	i-miss	d-miss	fills	miss stall
nofib-hartel										
Wang	869	4	117	3 110	4 912	48 017	1 624	1 089	2 610	46 839
Typecheck	95 900	24 902	47 698	1 310 044	878 080	15 502 284	14 939	88 679	102 601	1 952 934
Event	2 215	7	358	12 438	8 672	613 124	1 142	3 274	4 360	78 084
Sched	108 703	13 230	71 101	1 872 979	1 138 048	8 508 749	2 048	14 729	16 619	308 082
gadt										
RoseTree	208	1	36	863	1 793	962	615	575	1 124	19 710
Queue	492	1	38	1 144	6 816	40 088	875	1 652	2 447	44 468
MergeSort	2 216	1	39	2 975	8 144	155 571	438	3 221	3 615	65 386
InsertSort	497	1	29	974	14 736	137 802	283	3 953	4 209	75 721
BTree	599	1	44	1 380	2 048	33 307	530	1 329	1 825	33 763
Zipper	533	2	49	1 398	2 000	13 578	755	987	1 666	30 087
HeapSort	593	2	34	1 178	14 144	111 670	771	3 299	3 999	72 510
QuadTree	3 112	4	117	5 294	5 584	252 932	1 207	2 694	3 787	68 356
Patricia	1 105	2	67	2 280	2 544	40 627	704	727	1 380	24 482
List	5	1	5	114	129	1	1	3	3	58
Tuple	90	1	36	748	1 953	1 122	344	64	373	6 369
nofib-gc										
Power	68 382	13	17 494	415 565	288 192	1 804 559	1 364	6 149	7 404	134 167
Circsim	148 662	42 783	161 864	4 610 922	2 633 744	10 009 443	9 623	47 334	56 443	1 064 057

TABLE III: Full breakdown behind Table III of the letter, three-compiler execution time (compute cycles). Ratios are relative to GCC.

Program	GCC -O3	GHC	MicroHs	GHC/GCC	MicroHs/GCC
shootout					
nbody	948k	28.84M	5.05M	30.42	5.33
matrix	215k	25.36M	8.42M	117.95	39.16
sieve	286k	24.83M	5.85M	86.82	20.45
binarytrees	11.28M	22.06M	8.48M	1.96	0.75
fannkuchredux	99k	17.58M	4.48M	177.58	45.25
takfp	2.07M	15.99M	6.00M	7.72	2.90
methcall	1.17M	11.37M	13.54M	9.72	11.57
magicsquares	256k	8.42M	2.13M	32.89	8.32
nsievebits	40.5k	585k	311k	14.44	7.68
strcat	313k	7.17M	1.09M	22.91	3.48
objinst	6.02M	4.96M	4.80M	0.82	0.80
ary	32.1k	8.34M	2.89M	259.81	90.03
spectralnorm	242k	3.69M	1.81M	15.25	7.48
mandelbrot	2.16M	3.03M	2.61M	1.40	1.21
alloc	2.93M	2.71M	9.56M	0.92	3.26
spec.norm-pure	324k	2.40M	1.52M	7.41	4.69
nestedloop	7187	1.96M	8.71M	272.71	1211.91
heapsort	30k	1.60M	419k	53.33	13.97
hash	230k	1.56M	2.70M	6.78	11.74
wc	250k	1.20M	406k	4.80	1.62
random	767k	941k	6.92M	1.23	9.02
nsieve-pure	116k	232k	58k	2.00	0.50
nsieve	8120	115k	33k	14.16	4.06
rfib	7513	84k	68k	11.18	9.05
fasta	832k	2.63M	1.23M	3.16	1.48
harmonic	11.8k	941k	1.29M	79.75	109.32
partialsums	1.50M	6.23M	16.41M	4.15	10.94
others					
floydwarshall	24k	2.36M	1.01M	98.33	42.08
crc32	112k	1.77M	1.70M	15.80	15.18
fft	176k	1.16M	264k	6.59	1.50
heapsort (heap)	214k	509k	207k	2.38	0.97
tak	9052	20k	23k	2.21	2.54
gadt					
quadtree	372k	1.26M	348k	3.39	0.94
btree	300k	473k	84k	1.58	0.28
zipper	207k	379k	60k	1.83	0.29
patricia	136k	303k	102k	2.23	0.75
rosetree	144k	249k	34k	1.73	0.24

TABLE IV: Full PMC breakdown for the nofib programs ported and measured after the letter was written;

Program	arm	issue			cycle split			stalls / branch				d-cache		misses and RAM				
		cyc	instret	IPC	resid	freeze	bubble	starve	stall ld	misp T	misp NT	fill	wb	i-miss	d-miss	RAM rd	RAM wr	miss stall
Banner	mhs	174 601	77 355	0.44	3 082	10 281	83 883	9 513	53	44	45	288	4 783	792	1	27 472	95 897	7 013
	ghc	1 216 094	1 018 518	0.84	0	32 073	165 503	12 566	85 925	614	476	1 978	71	683	1 712	37 375	1 264	41 994
Boyer	mhs	11 484 290	5 711 857	0.50	645 499	3 881 903	1 245 031	956 305	116 148	52 219	62 294	10 297	450 393	66 254	7	165 968	7 173 040	385 284
	ghc	21 508 198	18 014 071	0.84	35	500 220	2 993 981	156 159	1 573 301	30 593	20 391	27 778	18 057	2 211	26 077	450 287	289 072	527 522
Compress2	mhs	5 121 874	1 831 330	0.36	90 495	2 478 984	721 065	602 232	36 993	2 793	8 005	11 596	257 529	53 640	89	219 552	4 054 559	337 719
	ghc	7 005 008	5 792 389	0.83	21	239 421	973 272	78 746	564 168	9 890	5 210	14 233	4 234	2 431	12 295	233 679	68 288	267 039
DigitsE1	mhs	392 776	243 305	0.62	5 361	60 328	83 782	3 668	13 133	810	570	104	17 562	3 472	2	2 384	280 815	5 868
	ghc	2 607 235	2 032 041	0.78	0	26 461	548 733	44 384	345 225	3 723	2 777	1 813	138	983	1 296	34 815	2 240	38 691
Eliza	mhs	18 470 338	9 840 012	0.53	564 792	6 242 050	1 823 484	1 430 662	256 487	33 372	28 589	12 797	806 329	104 771	39	249 598	12 883 510	456 174
	ghc	50 552 154	41 180 108	0.81	35	1 221 347	8 150 759	495 225	4 862 406	60 049	29 525	69 309	48 152	7 542	62 456	1 114 719	770 608	1 313 250
Fulsom	mhs	494 733	174 766	0.35	6 224	230 831	82 912	5 090	1 649	225	85	511	24 134	4 401	3	9 264	381 256	15 434
	ghc	726 271	566 598	0.78	0	35 094	124 579	30 314	46 786	2 425	2 469	2 772	199	2 244	989	50 175	3 344	55 864
Lift	mhs	1 439 831	533 468	0.37	22 651	503 976	379 736	337 611	10 056	529	972	5 820	66 216	12 263	10	132 144	1 067 077	188 098
	ghc	5 560 162	4 503 176	0.81	0	179 224	877 762	113 271	440 012	8 964	11 607	9 192	1 545	3 520	6 265	152 751	24 944	173 625
Maillist	mhs	31 132 701	15 020 473	0.48	4 226 402	4 412 125	7 473 701	4 872 077	239 813	70 681	77 224	70 930	782 803	164 904	11	2 908 464	10 724 464	2 239 334
	ghc	30 783 506	26 427 180	0.86	21	540 238	3 816 162	298 267	1 550 846	59 948	28 525	29 316	16 728	1 257	28 515	474 943	267 824	554 648
Mkhprog	mhs	5 343 131	1 880 849	0.35	41 260	814 576	2 606 446	2 077 353	73 360	241	4 090	43 563	185 676	37 717	8	1 064 577	3 329 168	1 052 243
	ghc	21 865 711	18 193 366	0.83	21	686 351	2 986 068	159 431	1 453 629	17 436	5 904	37 732	27 302	2 982	35 355	609 503	437 136	716 475
ParQueens	mhs	9 974 630	5 581 382	0.56	209 080	2 921 390	1 262 778	947 879	229 312	23 874	14 464	6 058	471 761	52 299	19	97 647	7 543 698	235 641
	ghc	10 414 678	9 083 912	0.87	0	58 038	1 272 728	90 653	566 461	7 582	16 644	3 383	76	645	3 156	59 887	1 264	67 310
Parfib	mhs	1 528 390	924 477	0.60	23 036	344 209	236 668	132 602	82 750	7 058	11 563	48	76 239	1 868	9	1 088	1 219 398	3 136
	ghc	1 278 322	1 059 073	0.83	0	407	218 842	31 946	66 441	8 888	7 153	36	0	367	69	6 383	64	6 981
Partak	mhs	8 179 666	3 424 981	0.42	86 630	3 941 638	726 417	589 587	113 463	16 454	7 463	1 537	410 728	55 576	40	25 247	6 564 607	103 344
	ghc	4 042 221	3 228 557	0.80	0	314	813 350	32 050	429 374	7 087	9 827	32	0	368	69	6 287	16	6 854
Sumeuler	mhs	8 377 834	4 288 467	0.51	127 787	292 069	3 669 511	1 960 509	258 195	14 137	32 876	967	368 408	16 090	10	58 847	5 932 924	42 857
	ghc	5 294 018	2 295 286	0.43	0	5 781	2 992 951	22 842	113 351	22 326	5 053	449	10	562	309	12 975	256	14 455

TABLE V: Python on the same core. Thirty-one programs from micropython’s tests/perf_bench, from pyperformance and from the Computer Language Shootout, under CircuitPython 10.3.0 in three tiers, beside MicroHs running the same program where the suite has it.

Program	problem size	CircuitPython (cycles)			MicroHs cycles	ratio		
		bytecode	native	viper		byte/viper	byte/mhs	viper/mhs
MicroHs runs the same program								
Mandel (nofib)	6 × 6, 75	3 031 965	2 937 029	661 341	394 335	4.58	7.69	1.68
fannkuchredux	n = 6	9 533 657	9 730 651	440 177	3 772 864	21.66	2.53	0.12
fannkuch (perf_bench) ^a	n = 6	12 088 640	12 027 257	427 100	3 772 864	28.30	3.20	0.11
nqueens (perf_bench)	n = 5	5 074 645	5 044 398	117 640	53 041	43.14	95.67	2.22
queens	n = 7	23 162 238	15 876 785	7 088 614	2 712 840	3.27	8.54	2.61
spectralnorm	n = 8	19 660 582	17 059 627	3 545 267	1 202 559	5.55	16.35	2.95
nbody	100 steps	25 415 627	26 680 743	1 240 974	3 612 169	20.48	7.04	0.34
mandelbrot	n = 12	13 230 978	12 904 707	1 313 044	1 749 575	10.08	7.56	0.75
nestedloop	n = 6	15 053 234	9 203 928	1 239 970	4 976 549	12.14	3.02	0.25
takfp	n = 6	118 594 163	100 946 859	180 525 886	4 302 842	0.66	27.56	41.96
random	20 000	80 433 752	71 706 438	1 449 884	4 356 074	55.48	18.46	0.33
binarytrees	n = 8	466 457 459	535 749 218	414 398 712	5 282 077	1.13	88.31	78.45
tak	9, 6, 3	376 961	276 494	162 359	14 198	2.32	26.55	11.44
rfib	n = 11	699 105	631 453	411 447	47 667	1.70	14.67	8.63
nsieve (list)	m = 100	648 223	523 553	651 795	38 466	0.99	16.85	16.94
fft	n = 32	1 421 690	1 273 561	527 855	209 038	2.69	6.80	2.53
heapsort	n = 100	1 490 929	946 393	120 029	318 989	12.42	4.67	0.38
floydwarshall	n = 12	2 316 894	1 524 142	197 112	663 257	11.75	3.49	0.30
crc32	n = 2000	26 082 366	24 969 143	395 727	1 164 737	65.91	22.39	0.34
alloc	20 × 1000	34 792 456	27 997 558	26 173 753	8 125 127	1.33	4.28	3.22
matrix	20 × 20	33 100 344	24 087 119	2 046 354	12 401 927	16.18	2.67	0.17
sieve	4096, 3	24 066 712	14 538 396	1 304 466	–	18.45	–	–
the same problem, another module's formulation								
spectral_norm (pyperformance)	n = 8	12 612 749	11 223 485	–	1 202 559	–	10.49	–
nbody (pyperformance) ^c	100 iters	25 415 627	26 680 743	–	–	–	–	–
micropython tests/perf_bench only								
float	1, 150	5 880 681	5 694 706	2 088 826	–	2.82	–	–
fft	2, 128	13 376 613	11 263 556	8 927 741	–	1.50	–	–
pidigits	1, 35	5 550 649	5 657 222	7 604 428	–	0.73	–	–
pystone	80	5 341 396	4 880 161	4 812 147	–	1.11	–	–
raytrace	5 × 5, 2	21 635 991	19 730 658	16 300 519	–	1.33	–	–
mandel	20 × 20	18 255 006	19 454 610	2 235 831	–	8.16	–	–
wordcount	2	2 892 726	2 593 639	2 638 297	–	1.10	–	–
chaos	50 × 50, 50	19 570 305	19 073 680	43 454 350	–	0.45	–	–
hexiom	level 10	28 978 854	25 610 546	21 504 256	–	1.35	–	–

^a the module reports the maximum flip count (10) where MicroHs reports the checksum and the flips as one number (4910); the run is the same. ^c this module computes the force term differently from the shootout entry; the nbody row of the first group is that entry, ported.